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**LIGHTWEIGHT HYBRID INTRUSION DETECTION SYSTEM
FOR SMALL-SCALE NETWORKS**

PROJECT PROPOSAL

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1. General Information

Project Title: Lightweight Hybrid Intrusion Detection System for Small-scale Networks
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2. Introduction

In today's interconnected world, network security is a paramount concern. While large companies invest in sophisticated and enterprise solutions, small-scale networks - such as small offices, home offices and businesses - often lack the technical knowledge and financial resources to implement robust solutions for network security.

These small networks are vulnerable to threats, the commercial Intrusion Detection Systems (IDS) are often too complex and heavyweight for consumer grade hardware that this type of businesses use. The growing use of end-to-end encryption makes traditional signature and rule based systems ineffective, since they can no longer inspect packet payloads. This highlights a crucial research gap, the need for a lightweight solution that can smartly identify new threats and suspicious patterns within encrypted traffic without causing significant performance impact.

3. Problem Statement

Small-scale networks lack real-time, accessible and resource efficient tools to detect common and novel network intrusions, especially with encrypted traffic being so common nowadays. This leaves them vulnerable to attacks that can disrupt traffic and compromise sensitive data.

The primary challenge is to develop a hybrid system that combines the strengths of traditional rule-based detection with the intelligence of machine learning to provide meaningful security alerts without high cost or performance overhead.

4. Aims and Objectives

Aim:

To design, develop and evaluate a lightweight, hybrid, real-time network monitoring tool that detects suspicious traffic patterns, common and novel attacks on a small-scale network.

Objectives:

1. Research and select an appropriate, efficient method for capturing network traffic in real-time using common libraries (e.g. Wireshark, tcpdump, scapy, pcap).
2. Develop a packet processing and advanced feature extraction module to create statistical flow-based features (e.g. packet size, frequency, flow duration).
3. Design and implement a hybrid detection engine that combines:
 - a. A simple, efficient rule-based component for traditional attacks (e.g. port scanning).
 - b. An unsupervised machine learning model (e.g. Isolation Forest, Autoencoder) to detect novel anomalies and behavioural deviations.
4. Investigate and implement techniques for Encrypted Traffic Analysis (ETA) using traffic metadata and flow statistics to identify suspicious activity patterns without decryption.
5. Create an interactive web-based dashboard to visualize the output of all our modules.
6. Evaluate the systems effectiveness by analysing its detection accuracy on a controlled test network and benchmarking its performance against public IDS datasets like CIC-IDS-2017/18

5. Scope

In Scope	Out of Scope
• The system will monitor traffic on a single network interface. • Detection will be a hybrid model, focusing on network-level anomalies (port scans, floods) and behavioral anomalies identified by the ML model. • Analysis of encrypted traffic metadata (e.g., flow data, packet size/timing) to infer activity. • The system will be a detection system, not a prevention system (it will alert, not block). • The dashboard will be an interactive visualization layer, not a full-featured SIEM.	• Deep Packet Inspection (DPI) for malicious payloads. • Direct decryption of encrypted traffic. • Supervised machine learning models that require a fully labeled, real-time dataset (focus is on unsupervised detection). • Automated incident response or traffic blocking.

6. Methodology and Approach

The project follows an incremental, phase-based development approach (subject to refinement during implementation)

Phase 1: Traffic Capture & Feature Extraction

Implement real-time packet capture using Python libraries (Scapy/pcapy/tcpdump). Extract flow-based statistical features (packet size, frequency, duration, protocol distribution) suitable for encrypted traffic analysis.

Phase 2: Hybrid Detection Engine

- Tier 1 (Rule-based): Real-time fast detection of known attacks (port scans, DoS, connection floods)
- Tier 2 (ML-based): Unsupervised anomaly detection for more novel attacks (Contingent on Tier 1 stability and consumer hardware performance testing)

Phase 3: Visualization

Integrate a front-end solution to the project for real-time monitoring and alerts. Most likely using Grafana/InfluxDB to cut development time from the front-end.

Phase 4: Evaluation

Test on CIC-IDS-2017/2018 datasets, measuring detection accuracy (TPR, FPR, F1-score) and system performance (CPU, RAM, latency).

Technology Stack (Not final)

Python 3.11+, NumPy/Pandas, scikit-learn, Grafana, InfluxDB, Docker Compose

Development Strategy

Build MVP with Python, implement Tier 1 first, add Tier 2 only if resource-efficient on consumer hardware. This plan will be adapted based on testing results and technical constraints encountered during development.

7. Expected Outcomes

The primary outcomes of this project will be:

1. A functional prototype of a lightweight, hybrid intrusion detection system.
2. A demonstrated ability to detect both known attacks (via rules) in real-time and novel anomalies (via ML) in near-real-time.
3. A working module capable of identifying suspicious patterns in encrypted traffic using metadata analysis.
4. An interactive web-based dashboard for visualizing network security alerts.
5. A comprehensive evaluation report benchmarking the system's performance and detection accuracy against public datasets.

8. Timeline

Phase	Description	Duration
Research & Design	Environment setup, library selection, system architecture design, dataset acquisition	Week 1-3
Development	Traffic capture, feature extraction, rule-based detection (Tier 1), dashboard integration	Week 4-8
Enhancement & Integration	Grafana/InfluxDB setup, ML module (Tier 2, if feasible), system integration	Week 9-11
Testing & Documentation	Dataset evaluation, performance benchmarking, final report	Week 12-14